MIL-F-3268D
11 February 1983
SUPERSEDING
MIL-F-3268C
9 February 1977

MILITARY SPECIFICATION

FITTINGS; PIPE, STEEL, SPECIAL (FOR TANKER UNLOADING INSTALLATIONS)

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers special fabricated steel pipe spool pieces and 90° elbows for use in tanker unloading operations.

1.2 Classification. Spool pieces and elbows covered by this specification shall be designated by a definitive military specification part number in the following form (see 6.2 and 6.5):

	M3268	XX	X
Military specification designation			
Kind and nominal pipe size (NPS)			
Code number (see 1.2.1)			
Gasket code letter (see 1.2.2)			

1.2.1 Kind and size of fitting. The kind and NPS of fitting (see 3.3, 6.2, and 6.6) are identified by a two digit number (see table I).

Table I. Code number to kind and size.

Kind of fitting	Fitting pipe size		
	6 NPS	8 NPS	12 NPS
Spool piece	01	02	03
Spool piece and elbow	04	05	06
Spool piece with tap-in	07	08	09
Spool piece with tap-in and elbow	10	11	12
Elbow	13	14	15

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commanding Officer (Code 156), Naval Construction Battalion Center, Port Hueneme, CA 93043, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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1.2.2 Gasket. The gasket requirement (see 3.3.4) is identified by a single alphabetic character (see table II).

Table II. Code letter to gasket requirement.

Code letter	Gasket requirement
Y	Gaskets furnished
N	Gaskets not furnished

2. APPLICABLE DOCUMENTS

2.1 *Government documents*.

2.1.1 Specifications, standards, and handbooks. Unless otherwise specified, the following specifications, standards, and handbooks of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DODISS) specified in the solicitation, form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-V-3 - Valve, Fittings, and Flanges (Except for Systems Indicated Herein), Packaging of.
MIL-T-704 - Treatment and Painting of Materiel.
MIL-R-2765 - Rubber Sheet, Strip, Extruded, and Molded Shapes, Synthetic, Oil Resistant.

STANDARDS

FEDERAL

FED-STD-595 - Colors.

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.

(Copies of specifications, standards, handbooks, drawings, and publications required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.2 Other publications. The following document(s) form a part of this specification to the extent specified herein. The issues of the documents which are indicated as DoD adopted shall be the issue listed in the current DODISS and the supplement thereto, if applicable.

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AMERICAN NATIONAL STANDARDS INSTITUTE, INC. (ANSI)

B16.5 - Steel Pipe Flanges, Flanged Valves and Fittings, Including Ratings for Class 150, 300, 400, 600, 900, 1500, and 2000.
B16.9 - Factory-Made Wrought Steel Butt-Welding Fittings.

(Application for copies should be addresssed to the American National Standards Institute, Inc., 1430 Broadway, New York, NY 10018.)

AMERICAN PETROLEUM INSTITUTE (API)

API 1104 - Standard for Field Welding of Pipe Lines.

(Application for copies should be addressed to American Petroleum Institute, 2802 K Street, N.W., Washington, DC 20006.)

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

ASTM A53 - Pipe, Steel, Black and Hot-Dipped Zinc Coated Welded and Seamless.
ASTM A105 - Forgings, Carbon Steel, Hot Dipping Component.
ASTM A120 - Pipe, Steel, Black and Hot Dipped (Galvanized) Welded and Seamless, for Ordinary Uses.
ASTM A234 - Pipe Fittings of Wrought Carbon Steel and Alloy Steel for Moderate and Elevated Temperatures.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

(Industry association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 First article. When specified (see 6.2), the contractor shall furnish a sample unit for first article inspection and approval (see 4.3 and 6.4).

3.2 Materials. Materials used shall be free from defects which would adversely affect the performance or maintainability of individual components or of the overall assembly. Materials not specified herein shall be of the same quality used for the intended purpose in commercial practice. Unless otherwise specified herein, all equipment, material, and articles incorporated in the work covered by this specification are to be new and fabricated using materials produced from recovered materials to the maximum extent possible without jeopardizing the intended use. The term "recovered materials" means materials which have been collected or recovered from solid waste and reprocessed to become a source of raw materials, as opposed to virgin raw materials. None of the above shall be interpreted to mean that the use of used or rebuilt products are allowed under this specification unless otherwise specified.

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3.2.1 Steel. Butt-welded type fittings shall conform to ANSI B16.9, standard weight, material to ASTM A234. Standard weight steel pipe shall conform to ASTM A53 or ASTM A120, standard weight, black finish. Forged-steel-type pipe flanges and flanged fittings shall conform to ANSI B16.5, class 150, material to ASTM A105.

3.2.2 Gasket material. Gaskets shall be fabricated from material conforming to MIL-R-2765, not less than 1/16-inch thick. The material shall be tested in accordance with 4.4.1.

3.3 Design. The fitting(s) shall be fabricated as shown on figure 1. Each fabricated fitting shall withstand an internal hydrostatic pressure of 350 pounds per square inch (psi).

3.3.1 Spool piece. The spool piece shall include a standard weight steel pipe, approximately 2-feet long, with a slip-on flange welded to each end. Two lifting ears of plate steel spaced 180° apart as shown in figure 1, shall be welded longitudinally to the pipe body midway between the flange faces, and approximately in line with the midpoints between the pairs of bolt holes in each flange. The lifting ears shall be welded to the pipe with a 3/8-inch fillet weld all around.

3.3.2 Spool piece tap-in. When a spool piece with tap-in is required (see 1.2 and 6.2), the spool piece shall have two 1-1/4 inch outside diameter (OD) by 1/2-inch thick bosses welded 5-1/2 inches from one end as shown on figure 1. The two bosses shall be spaced 180° apart. A hole shall be drilled through the bosses and the pipe, and taped with 1/4-inch American Standard taper pipe thread (NPT). Fittings with 1/4-inch NPT threaded connections and contoured at the base may be used to provide threaded outlets in lieu of bosses upon approval of the contracting officer. Each threaded outlet shall be fitted with a 1/4-inch steel plug.

3.3.3 Elbow. The elbow shall include a welded neck flange; a standard weight, black steel, 90° long radius butt-welded elbow; a butt-welded lap-joint stud end; and a lap-joint flange loosely fitted to the stub end. A lifting ear of plate steel shall be welded longitudinally to the long radius of the 90° elbow, approximately midway between the ends, by a 3/8-inch fillet weld all around.

3.3.4 Gaskets. Unless otherwise specified (see 1.2 and 6.2), gaskets shall be furnished for all flanges except when spool piece (with or without tap-in) and elbow are required. When the spool piece and elbow are required and gaskets are to be furnished, one gasket for the elbow end, one gasket for the spool piece end, and one gasket between the mating ends of the spool piece and elbow shall be furnished. When the spool piece and elbow are required, and gaskets are not to be furnished, one gasket between the mating ends of the spool piece and elbow shall be furnished. The gaskets shall be precut and shall have the following inside diameter (ID) and OD for the specified NPS (see table III):

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Table III. Gasket size and tolerance.

NPS	Diameter - inches ID	Diameter - inches OD	Tolerance
6	6	8 5/8	-0 +1/16
8	8	10 7/8	-0 +1/16
12	12	16	0 +1/16

3.4 Identical items. All of the same classification furnished under a specific contract shall be physically and mechanically identical. Parts, accessories, assemblies, and components are included in this requirement. Written approval for deviations must be obtained in advance from the contracting officer.

3.5 Standard product. Except where modified herein, it is intended that the basic component parts of the fittings shall be regular commercial products of the manufacturer or his suppliers. The parts shall be new, unused, and free from defects or imperfections.

3.6 Identification marking. The manufacturer's name, initials, or trade mark shall be legibly cast or die-stamped on each fitting.

3.7 Cleaning, treatment, and painting. Exterior surfaces of fittings, except flange faces, shall be cleaned, treated, and painted in accordance with MIL-T-704, type B. The color of the finish coat shall be gloss black No. 17038 of FED-STD-595.

3.8 Workmanship.

3.8.1 Steel fabrication. The steel used in fabrication shall be free from kinks, sharp bends, and other conditions which would be deleterious to the finished product. Manufacturing processes shall not reduce the strength of the steel to a value less than intended by the design. Manufacturing processes shall be done neatly and accurately. All bends shall be made by controlled means to insure uniformity of size and shape.

3.8.2 Welding. Surface to be welded shall be free from foreign matter which would be injurious to the weld. Welding procedures shall be in accordance with API 1104. Welds shall be of sufficient size and shape to develop the full design strength of the parts connected by the welds. Welds shall transmit imposed stresses without permanent deformation or failure when subjected to proof or service loadings.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection require-

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ments specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.2.1).
b. Quality conformance inspection (see 4.2.2).

4.2.1 First article inspection. The first article inspection shall be performed on one fitting assembly when a first article is required (see 3.1 and 6.2). This inspection shall include the examination of 4.4 and the tests of 4.5.1 and 4.5.2. The first article may be either a first production item or a standard production item from the supplier's current inventory provided the item meets the requirements of the specification and is representative of the design, construction, and manufacturing technique applicable to the remaining items to be furnished under the contract.

4.2.2 Quality conformance inspection. The quality conformance inspection shall include the examination of 4.4, the tests of 4.5.2, and the packaging inspection of 4.6. This inspection shall be performed on the samples selected in accordance with 4.3.

4.3 Sampling. Sampling shall be in accordance with MIL-STD-105.

4.3.1 Sampling for examination. Examination shall be based on inspection level II and an Acceptable Quality Level (AQL) of 4.0 percent defective.

4.3.2 Sampling for tests. Tests shall be based on inspection level S-3 and an AQL of 1.5 percent defective.

4.4 Examination. Each sample shall be examined for compliance with the requirements specified in section 3 of this specification. Any redesign or modification of the contractor's standard product to comply with specified requirements, or any necessary redesign or modification following failure to meet specified requirements shall receive particular attention for adequacy and suitability. This element of inspection shall encompass all visual examinations and dimensional measurements. Noncompliance with any specified requirements or presence of one or more defects preventing or lessening maximum efficiency shall constitute cause for rejection.

4.5 Tests.

4.5.1 First article test. The gasket material shall be tested in accordance with MIL-R-2765.

4.5.2 Functional tests. Each sample selected in accordance with 4.4.2 shall be tested to determine compliance with this specification. Tests shall be conducted as specified in 4.5.2.1 and 4.5.2.2.

4.5.2.1 Hydrostatic test. Each sample fitting shall be subjected to an internal hydrostatic pressure of 350 psi for 15 minutes. Any evidence of leakage shall be considered a defect and the fitting shall be rejected.

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4.5.2.2 Welding test. Radiographic inspection procedures and tests should be conducted on the butt-welds of elbow fittings in accordance with API 1104 to determine if the welds conform to the requirement of 3.8.2.

4.6 Packaging inspection. The preservation, packing, and marking of the item shall be inspected to verify conformance to the requirements of section 5.

5. PACKAGING

5.1 Preservation, packing, and marking. Preservation, packing, and marking shall be in accordance with the requirements of MIL-V-3 with the level of preservation and the level of packing as specified (see 6.2).

6. NOTES

6.1 Intended use. The fittings are intended for use with submerged loading line installations in conjunction with tanker unloading operations of liquid fuels. The spool piece is used for hoisting and securing the hose which connects the submerged pipeline to the ship receiving or discharging fuel.

6.2 Ordering data. Acquisition documents should specify the following:

a. Title, number, and date of this specification.
b. Military specification part number required (see 1.2, 1.2.1, 3.3.2, 3.3.4, and 6.5).
c. When a first article is required for inspection and approval (see 3.1, 4.2.1, and 6.4).
d. Level of preservation and packing required (see 5.1).

6.3 Data requirements. When this specification is used in an acquisition which incorporates a DD Form 1423 and invokes the provisions of paragraph 7-104.9(n) of the Defense Acquisition Regulations (DAR), the data requirements will be developed as specified by an approved Data Item Description (DD Form 1664) and delivered in accordance with the Contract Data Requirements List (DD Form 1423) incorporated into the contract. When the provisions of DAR 7-104.9(n) are not invoked, the data shall be delivered in accordance with the contract requirements.

6.4 First article. When a first article inspection is required, the item will be tested and should be a first article sample, or it may be a standard production item from the contractor's current inventory as specified in 4.2.1. The first article should consist of one unit. The contracting officer should include specific instructions in acquisition documents regarding arrangements for examination, test, and approval of the first article.

6.5 Definitive military specification part number. The military specification part number is a definitive part number which corresponds to the kind and size of special fittings covered by this specification and defines the requirement of the options presented under this specification. The military specification part number is formed by combining the military specification code number (M3268 for MIL-F-3268) with a dash after it, the kind and size code number, and the gasket code letter as shown in 1.2.

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6.6 The dimensionless designator NPS (nominal pipe size) has been substituted in this specification for such traditional terms as "nominal diameter", "size", and "nominal size".

6.7 Changes from previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue, due to the extensiveness of the changes.

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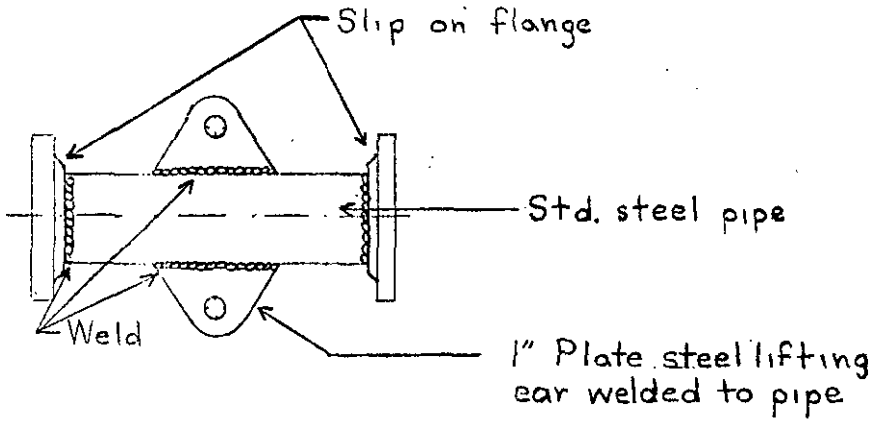
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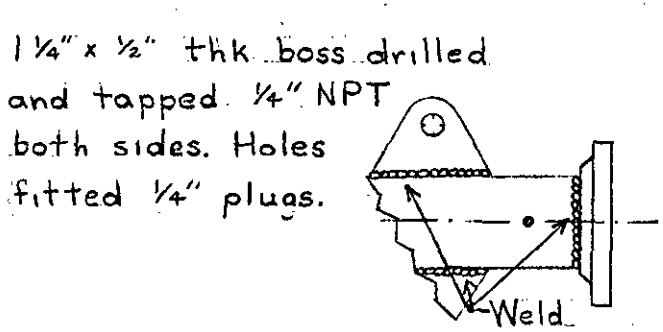
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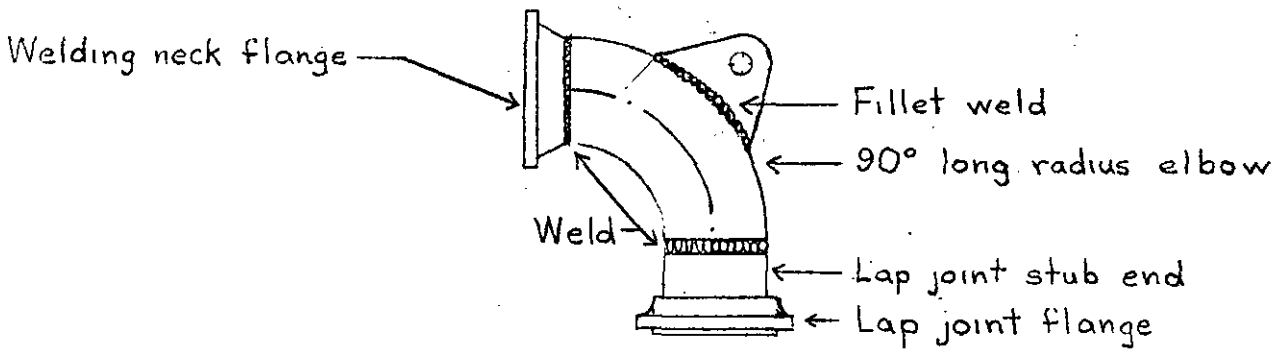


A. SPOOL PIECE



NOTE: All other dimensions same as figure A.

B. SPOOL PIECE WITH TAP-IN



C. ELBOW

FIGURE 1. SPECIAL STEEL PIPE FITTINGS.

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